

# **Use Case: data-driven camera calibration for light field camera systems**

**Author:** Youssef Daoudi El Boukhrissi (4277022)

4. Januar 2026

# Use Case: Cross-Shaped Camera Array

## [Center + North/East/South/West]

### Phase 1 — Initial Lab Calibration (once)

- **Target-based, OpenCV:** Checkerboard-based initialization to estimate intrinsic/extrinsic parameters.
- **Global refinement:** Bundle adjustment over the collected images points to improve estimated parameters and obtain a consistent calibration.
- **Baseline:** Store the resulting calibration as the known-good reference for later comparison.

### Phase 2 — Health-Check (periodic)

- **Trigger:** Fixed interval (e.g. every 3 minutes), temperature changes, or system restart.
- **Small image set (sampling):** Periodically capture a small, recent image set during normal operation.
- **Parameter-based health metrics:**
  - *Intrinsic deviation:* Relative deviation of key intrinsic parameters (e.g. principal point, effective focal length, low-order distortion) compared to the reference (initial) calibration.
  - *Extrinsic deviation:* Change of estimated extrinsic parameters relative to the reference calibration.
  - *Parameter drift:* Normalized parameter change between consecutive health-check windows.
- **Health score:** Combine the normalized intrinsic and extrinsic deviations into a single health score.
- **Decision:** If the health score exceeds a predefined threshold for  $\geq 2$  consecutive health-check windows, the calibration is considered degraded and Phase 3 is triggered.

### Phase 3 — Target-Free Online Recalibration

- **Larger image set:** Collect a larger image set from recent normal operation to increase parameter stability.
- **Re-estimation scope:**
  - Re-estimate intrinsic and extrinsic calibration parameters using the larger image set.
  - Keep structural properties of the camera rig (e.g. baseline and physical layout) fixed.

- **Method:** Perform a LiFCal-based parameter estimation on the extended image set.
- **Commit rule:** The updated calibration is adopted only if the resulting parameter deviations and health score improve compared to the previous calibration; otherwise, the update is discarded and retried later.